DAILY PRACTICE CALENDAR

Algebra 1 Competition Math (Grade 7) • 2027–2028

34 weeks • Monday–Friday • 10 problems a day (about 30–40 minutes)

Experience first. Symbols second. Understanding always.

How to Use This Calendar

- **Mon–Thu:** 10 problems on the day's skill: **8 new + 2 review** from earlier weeks. Start with the **Try** problem as the warm-up.
- **Friday:** Weekly Quiz of 10 mixed problems (8 this week + 2 review). Copy every missed problem into the **error log** and redo it next Monday.
- **Holidays and breaks** are marked. A holiday's skill moves to the weekend; break weeks are not counted in the 34 weeks.
- This course follows Pre-Algebra Competition Math: it sharpens **AMC 8 and MATHCOUNTS** skills and gives a first look at **AMC 10**. Answers to every **Try** problem are on the last page.

Year at a Glance

Wk	Dates	Topic
1	Aug 16–Aug 20	Problem-Solving Toolkit for Algebra
2	Aug 23–Aug 27	Multi-Step & Literal Equations
3	Aug 30–Sep 3	Absolute Value
4	Sep 6–Sep 10	Rate, Mixture & Work Problems
5	Sep 13–Sep 17	Slope & Linear Functions
6	Sep 20–Sep 24	Equations of Lines
7	Sep 27–Oct 1	Systems of Equations
8	Oct 4–Oct 8	Systems Word Problems
9	Oct 11–Oct 15	Inequalities & Integer Solutions
10	Oct 18–Oct 22	Review & Mini-Contest 1
11	Oct 25–Oct 29	Exponent Rules
12	Nov 1–Nov 5	Radicals
13	Nov 8–Nov 12	Big Numbers & Scientific Notation
14	Nov 15–Nov 19	Number Theory with Algebra
	Nov 22–Nov 26	*Thanksgiving Break*
15	Nov 29–Dec 3	Remainders & Modular Arithmetic
16	Dec 6–Dec 10	Polynomials & Special Products
17	Dec 13–Dec 17	Factoring
	Dec 20–Dec 24	*Winter Break*
	Dec 27–Dec 31	*Winter Break*
18	Jan 3–Jan 7	Solving Quadratics
19	Jan 10–Jan 14	Vieta's Formulas & the Discriminant
20	Jan 17–Jan 21	Mid-Year Mock Contest & Review
21	Jan 24–Jan 28	Functions & Function Notation
22	Jan 31–Feb 4	Arithmetic & Geometric Sequences
23	Feb 7–Feb 11	Series & Clever Sums
24	Feb 14–Feb 18	Coordinate Geometry
25	Feb 21–Feb 25	Similar & Right Triangles
26	Feb 28–Mar 3	Area, Circles & Algebra
27	Mar 6–Mar 10	Counting Strategies
28	Mar 13–Mar 17	Combinations & Pascal's Triangle
29	Mar 20–Mar 24	Probability
	Mar 27–Mar 31	*Spring Break*
30	Apr 3–Apr 7	Expected Value & Statistics
31	Apr 10–Apr 14	Logic, Parity & Pigeonhole
32	Apr 17–Apr 21	Contest Strategy: AMC 8 & AMC 10 Preview
33	Apr 24–Apr 28	Full Mock Contest & Review
34	May 1–May 5	Final Contest & Celebration

Units 1 & 6 · Unit 2 · Unit 3 · Unit 4 · Unit 5 · Wrap-Up · Break

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WEEK 1 • Aug 16–Aug 20, 2027

Unit 1: Equations, Inequalities & Absolute Value

Problem-Solving Toolkit for Algebra

Core: Translate words into equations, then check the answer in the story.

Mon Aug 16

Words to equations

TRY Five more than twice a number is 37. Find the number.

Tue Aug 17

Two clues, two numbers

TRY Two numbers add to 30 and differ by 8. Find the larger number.

Wed Aug 18

Draw a picture

TRY A rectangle's length is 3 more than twice its width. Its perimeter is 54. Find the width.

Thu Aug 19

Work backward

TRY Lily spent half her money, then $6, then half of what was left. She had $9 left. How much did she start with?

Fri Aug 20

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 2 • Aug 23–Aug 27, 2027

Unit 1: Equations, Inequalities & Absolute Value

Multi-Step & Literal Equations

Core: Clear fractions, distribute, collect like terms — and solve for any letter.

Mon Aug 23

Distribute & collect

TRY Solve $4(x-3)-2(x+5)=6$.

Tue Aug 24

Clearing fractions

TRY Solve $\frac{x+1}{3}-\frac{x-2}{4}=3$.

Wed Aug 25

Literal equations

TRY Solve $A=\frac{1}{2}h(b+c)$ for b.

Thu Aug 26

Infinitely many solutions

TRY For what value of k does $3(x+2)=3x+k$ have infinitely many solutions?

Fri Aug 27

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 3 • Aug 30–Sep 3, 2027

Unit 1: Equations, Inequalities & Absolute Value

Absolute Value

Core: $|x-a|$ is the distance from x to a on the number line.

Mon Aug 30

Absolute value equations

TRY Solve $|2x-5|=9$.

Tue Aug 31

Absolute value inequalities

TRY Solve $|x-3|<4$.

Wed Sep 1

Counting integer solutions

TRY How many integers x satisfy $|x+2|\le 5$?

Thu Sep 2

Sums of distances

TRY Find the smallest possible value of $|x-1|+|x-7|$.

Fri Sep 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 4 • Sep 6–Sep 10, 2027

Unit 1: Equations, Inequalities & Absolute Value

Rate, Mixture & Work Problems

Core: Organize every story in a table: rate $\times$ time $=$ amount.

Mon Sep 6

HOLIDAY: Labor Day

No class. Do this skill's 10 problems on the weekend:

Distance, rate & time

TRY Two trains leave a station in opposite directions at 50 mph and 70 mph. When are they 300 miles apart?

Tue Sep 7

Mixtures

TRY How many liters of 60% acid must be added to 10 L of 20% acid to make 30% acid?

Wed Sep 8

Filling & draining

TRY One pipe fills a tank in 4 hours; another empties it in 6 hours. With both open, how long to fill it?

Thu Sep 9

Catch-up problems

TRY Ann starts walking at 9:00 at 4 km/h. Ben starts from the same place at 10:00, biking 12 km/h. When does Ben catch up?

Fri Sep 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 5 • Sep 13–Sep 17, 2027

Unit 2: Linear Functions & Systems

Slope & Linear Functions

Core: Slope is rise over run: a constant rate of change.

Mon Sep 13

Slope from two points

TRY Find the slope of the line through $(-2, 7)$ and $(4, -5)$.

Tue Sep 14

Intercepts

TRY Find the x- and y-intercepts of $y = -3x + 6$.

Wed Sep 15

Collinear points

TRY For what k are $(1, 3)$, $(4, 9)$ and $(6, k)$ on one line?

Thu Sep 16

Rates in context

TRY A plumber charges $140 for 2 hours and $230 for 5 hours (a linear rule). Find the hourly rate and the fixed fee.

Fri Sep 17

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 6 • Sep 20–Sep 24, 2027

Unit 2: Linear Functions & Systems

Equations of Lines

Core: A point and a slope determine a line.

Mon Sep 20

Point–slope form

TRY Write the line through $(3, -1)$ with slope 2 in the form $y = mx + b$.

Tue Sep 21

Perpendicular lines

TRY Find the line through $(4, 3)$ perpendicular to $y = 2x + 1$.

Wed Sep 22

Intercepts & area

TRY Find the area of the triangle formed by $3x + 4y = 24$ and the two axes.

Thu Sep 23

Standard form

TRY Write the line through $(0, 4)$ and $(6, 0)$ as $Ax + By = C$ with integers.

Fri Sep 24

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 7 • Sep 27–Oct 1, 2027

Unit 2: Linear Functions & Systems

Systems of Equations

Core: Substitute or eliminate to find where two lines meet.

Mon Sep 27

Substitution

TRY Solve $y = 2x - 1$ and $3x + y = 19$.

Tue Sep 28

Elimination

TRY Solve $2x + 3y = 12$ and $4x - 3y = 6$.

Wed Sep 29

Shortcut systems

TRY If $x + y = 10$ and $x - y = 4$, find $x \cdot y$.

Thu Sep 30

No solution

TRY For what a does the system $2x + ay = 5$, $4x + 6y = 7$ have no solution?

Fri Oct 1

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 8 • Oct 4–Oct 8, 2027

Unit 2: Linear Functions & Systems

Systems Word Problems

Core: Two unknowns need two equations.

Mon Oct 4

Digit problems

TRY A two-digit number's digits add to 11. Reversing the digits makes it 27 larger. Find the number.

Tue Oct 5

Scoring problems

TRY A 30-question test gives $+4$ for right and -1 for wrong. Sam answered all and scored 85. How many were right?

Wed Oct 6

Current problems

TRY A boat goes 24 km downstream in 2 h and back upstream in 3 h. Find the speed of the current.

Thu Oct 7

Three unknowns

TRY If $x + y = 7$, $y + z = 9$ and $x + z = 10$, find $x + y + z$.

Fri Oct 8

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 9 • Oct 11–Oct 15, 2027 — Unit 2: Linear Functions & Systems

Inequalities & Integer Solutions **Core:** Describe a region, then count the integer points inside it.

Mon Oct 11
Compound inequalities
TRY Solve $-3 < 2x + 1 \leq 9$.

Tue Oct 12
Integer solutions
TRY How many pairs of positive integers (x, y) satisfy $2x + 3y = 30$?

Wed Oct 13
Lattice points
TRY How many points with integer coordinates lie on the segment from $(0, 0)$ to $(12, 18)$, endpoints included?

Thu Oct 14
Counting regions
TRY How many points (x, y) with positive integer coordinates satisfy $x + y \leq 6$?

Fri Oct 15
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 10 • Oct 18–Oct 22, 2027 — Unit 2: Linear Functions & Systems

Review & Mini-Contest 1 **Core:** Pull Units 1–2 together and practice contest habits.

Mon Oct 18
Review: absolute value
TRY Solve $|x - 4| = 2x + 1$.

Tue Oct 19
Review: lines
TRY Find the slope of $5x - 2y = 10$.

Wed Oct 20
Review: systems
TRY Adult tickets cost \$8 and child tickets \$5. 50 tickets sold for \$310. How many adult tickets?

Thu Oct 21
Timed set: 10 problems in 20 min
TRY A line passes through $(2, 5)$ and $(6, 13)$. Find its y-intercept.

Fri Oct 22
Mini-Contest 1
Units 1–2, timed. Afterward, start an error log of every missed problem.

WEEK 11 • Oct 25–Oct 29, 2027 — Unit 3: Exponents, Radicals & Number Theory

Exponent Rules **Core:** Same base: add exponents to multiply, subtract to divide.

Mon Oct 25
Product & quotient rules
TRY Simplify $\dfrac{x^5 \cdot x^3}{x^4}$.

Tue Oct 26
Zero & negative exponents
TRY Compute $2^{-3} + 4^0 \times 3^{-1}$.

Wed Oct 27
Rewrite with the same base
TRY Solve $4^x = 8^3$.

Thu Oct 28
Comparing powers
TRY Which is larger: 2^{100} or 3^{60}?

Fri Oct 29
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 12 • Nov 1–Nov 5, 2027 — Unit 3: Exponents, Radicals & Number Theory

Radicals **Core:** Pull out perfect squares; rationalize denominators.

Mon Nov 1
Simplifying radicals
TRY Simplify $\sqrt{180}$.

Tue Nov 2
Adding radicals
TRY Simplify $\sqrt{50} + \sqrt{18} - \sqrt{8}$.

Wed Nov 3
Conjugates
TRY Compute $(2 + \sqrt{3})(2 - \sqrt{3})$.

Thu Nov 4
Rationalizing
TRY Simplify $\dfrac{6}{\sqrt{3}}$.

Fri Nov 5
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 13 • Nov 8–Nov 12, 2027 — Unit 3: Exponents, Radicals & Number Theory

Big Numbers & Scientific Notation **Core:** Powers of 10 tell you the size of a number.

Mon Nov 8
Scientific notation
TRY Write $(3 \times 10^5)(4 \times 10^{-2})$ in scientific notation.

Tue Nov 9
Counting digits
TRY How many digits does $2^{10} \times 5^{12}$ have?

Wed Nov 10
Trailing zeros
TRY How many zeros are at the end of $30!$?

Thu Nov 11
HOLIDAY: Veterans Day
No class. Do this skill's 10 problems on the weekend:
Estimating roots
TRY Between which two consecutive integers is $\sqrt{150}$?

Fri Nov 12
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 14 • Nov 15–Nov 19, 2027 — Unit 3: Exponents, Radicals & Number Theory

Number Theory with Algebra **Core:** Factor expressions to reveal facts about numbers.

Mon Nov 15
Difference of squares
TRY Compute $101^2 - 99^2$.

Tue Nov 16
Parity of expressions
TRY For how many integers n from 1 to 20 is $n^2 + n$ odd?

Wed Nov 17
Number of divisors
TRY How many positive divisors does $2^3 \times 3^2 \times 5$ have?

Thu Nov 18
GCF $\times$ LCM
TRY Two numbers have GCF 6 and LCM 180. One of them is 36. Find the other.

Fri Nov 19
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Thanksgiving Break • Nov 22–Nov 26, 2027 • no class (not counted in the 34 weeks)

WEEK 15 • Nov 29–Dec 3, 2027 — Unit 3: Exponents, Radicals & Number Theory

Remainders & Modular Arithmetic **Core:** Keep only the remainders: they add and multiply like ordinary numbers.

Mon Nov 29
Remainders of products
TRY What is the remainder when 2027×2028 is divided by 9?

Tue Nov 30
Units digits
TRY What is the units digit of 2^{2028}?

Wed Dec 1
Remainders of powers
TRY What is the remainder when 3^{100} is divided by 7?

Thu Dec 2
Two conditions
TRY Find the smallest positive integer that leaves remainder 2 when divided by 3 and remainder 3 when divided by 5.

Fri Dec 3
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 16 • Dec 6–Dec 10, 2027 — Unit 4: Polynomials & Quadratics

Polynomials & Special Products **Core:** Recognize the patterns: $(a+b)^2$ and $(a+b)(a-b)$.

Mon Dec 6
Multiplying polynomials
TRY Expand $(x+3)(2x-5)$.

Tue Dec 7
Perfect squares
TRY If $a+b=7$ and $a \cdot b = 10$, find a^2+b^2.

Wed Dec 8
Difference of squares
TRY If $x^2 - y^2 = 48$ and $x+y=12$, find $x-y$.

Thu Dec 9
Clever computation
TRY Compute 999×1001.

Fri Dec 10
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 17 • Dec 13–Dec 17, 2027 — Unit 4: Polynomials & Quadratics

Factoring — **Core:** Look for a common factor first, then patterns, then trinomials.

Mon Dec 13
Common factors
TRY Factor $6x^3 - 9x^2$.

Tue Dec 14
Trinomials
TRY Factor $x^2 - 5x - 24$.

Wed Dec 15
Leading coefficient $\neq 1$
TRY Factor $2x^2 + 7x + 3$.

Thu Dec 16
Factoring to count
TRY How many pairs of positive integers (x, y) satisfy $x^2 - y^2 = 15$?

Fri Dec 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Winter Break • Dec 20–Dec 24, 2027 • no class (not counted in the 34 weeks)

Winter Break • Dec 27–Dec 31, 2027 • no class (not counted in the 34 weeks)

WEEK 18 • Jan 3–Jan 7, 2028 — Unit 4: Polynomials & Quadratics

Solving Quadratics — **Core:** Factor, complete the square, or use the quadratic formula.

Mon Jan 3
Zero-product property
TRY Solve $x^2 - 7x + 12 = 0$.

Tue Jan 4
Completing the square
TRY Write $x^2 + 6x + 2$ in the form $(x + h)^2 + k$.

Wed Jan 5
Quadratic formula
TRY Solve $x^2 - 4x - 1 = 0$.

Thu Jan 6
Quadratic word problems
TRY A rectangle has area 60, and its length is 4 more than its width. Find the width.

Fri Jan 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 19 • Jan 10–Jan 14, 2028 — Unit 4: Polynomials & Quadratics

Vieta's Formulas & the Discriminant — **Core:** For $x^2 + bx + c$: sum of roots $= -b$, product $= c$.

Mon Jan 10
Sum & product of roots
TRY Find the sum and product of the roots of $x^2 - 9x + 14 = 0$.

Tue Jan 11
Symmetric expressions
TRY If r and s are the roots of $x^2 - 5x + 3 = 0$, find $r^2 + s^2$.

Wed Jan 12
The discriminant
TRY For what positive k does $x^2 + kx + 9 = 0$ have exactly one solution?

Thu Jan 13
Building a quadratic
TRY Find b and c so that $x^2 + bx + c = 0$ has roots 3 and -7.

Fri Jan 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 20 • Jan 17–Jan 21, 2028 — Unit 4: Polynomials & Quadratics

Mid-Year Mock Contest & Review — **Core:** Review Units 1–4 and practice pacing a full contest.

Mon Jan 17
HOLIDAY: MLK Jr. Day
No class. Do this skill's 10 problems on the weekend:
Review: exponents & radicals
TRY Simplify $\sqrt{12} + \sqrt{27}$.

Tue Jan 18
Review: number theory
TRY What is the units digit of 7^{2028}?

Wed Jan 19
Review: quadratics
TRY Solve $x^2 = 5x + 14$.

Thu Jan 20
Pacing: skip & return
TRY 15-minute set. Warm-up: if $x + \frac{1}{x} = 5$, find $x^2 + \frac{1}{x^2}$.

Fri Jan 21
Mid-Year Mock Contest
Full-length, Units 1–4. Then sort errors by topic in the error log.

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WEEK 21 • Jan 24–Jan 28, 2028 — Unit 5: Functions, Sequences & Geometry

Functions & Function Notation

Core: A function turns each input into exactly one output.

Mon Jan 24
Evaluating functions
TRY If $f(x) = 2x^2 - 3x$, find $f(-2)$.

Tue Jan 25
Composition
TRY If $f(x) = x + 3$ and $g(x) = 2x$, find $f(g(5)) - g(f(5))$.

Wed Jan 26
Functional equations
TRY If $f(x+1) = 3x - 2$ for every x, find $f(5)$.

Thu Jan 27
Working backward
TRY If $f(x) = \frac{x-4}{3}$ and $f(x) = 7$, find x.

Fri Jan 28
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 22 • Jan 31–Feb 4, 2028 — Unit 5: Functions, Sequences & Geometry

Arithmetic & Geometric Sequences

Core: Arithmetic sequences add the same number; geometric ones multiply by it.

Mon Jan 31
Arithmetic nth term
TRY The 5th term of an arithmetic sequence is 23 and the 12th is 51. Find the first term.

Tue Feb 1
Geometric nth term
TRY Find the 7th term of 3, 6, 12, ...

Wed Feb 2
Missing terms
TRY Put two numbers between 2 and 54 to make a geometric sequence.

Thu Feb 3
Which kind?
TRY x, $2x + 2$, $3x + 3$ form a geometric sequence, and $x \neq -1$. Find x.

Fri Feb 4
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 23 • Feb 7–Feb 11, 2028 — Unit 5: Functions, Sequences & Geometry

Series & Clever Sums

Core: Pair terms, telescope, or use a formula.

Mon Feb 7
Arithmetic series
TRY Find $5 + 9 + 13 + \cdots + 81$.

Tue Feb 8
Geometric series
TRY Find $1 + 2 + 4 + \cdots + 2^{10}$.

Wed Feb 9
Telescoping
TRY Find $\frac{1}{1\times3} + \frac{1}{3\times5} + \cdots + \frac{1}{19\times21}$.

Thu Feb 10
Odd-number sums
TRY Find $1 + 3 + 5 + \cdots + 99$.

Fri Feb 11
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 24 • Feb 14–Feb 18, 2028 — Unit 5: Functions, Sequences & Geometry

Coordinate Geometry

Core: Distance, midpoint and slope turn pictures into algebra.

Mon Feb 14
Distance formula
TRY Find the distance between $(1, -2)$ and $(7, 6)$.

Tue Feb 15
Triangle areas
TRY Find the area of the triangle with vertices $(1, 1)$, $(7, 1)$ and $(4, 9)$.

Wed Feb 16
Midpoints & circles
TRY A circle has a diameter from $(-3, 4)$ to $(5, -2)$. Find its center and radius.

Thu Feb 17
Quadrilateral areas
TRY Find the area of the quadrilateral with vertices $(0, 0)$, $(6, 0)$, $(8, 5)$, $(2, 5)$.

Fri Feb 18
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 25 • Feb 21–Feb 25, 2028 — Unit 5: Functions, Sequences & Geometry

Similar & Right Triangles **Core:** Similar figures share ratios; right triangles share $a^2 + b^2 = c^2$.

Mon Feb 21

HOLIDAY: Presidents' Day

No class. Do this skill's 10 problems on the weekend:

Similar triangles

TRY A 6-ft person casts a 4-ft shadow. At the same time, a tree casts a 30-ft shadow. How tall is the tree?

Tue Feb 22

Pythagorean triples

TRY A right triangle has legs 20 and 48. Find the hypotenuse.

Wed Feb 23

Special right triangles

TRY A 30-60-90 triangle has hypotenuse 10. Find both legs.

Thu Feb 24

Altitude to the hypotenuse

TRY A right triangle has legs 6 and 8. Find the altitude to the hypotenuse.

Fri Feb 25

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 26 • Feb 28–Mar 3, 2028 — Unit 5: Functions, Sequences & Geometry

Area, Circles & Algebra **Core:** Turn a picture into an equation, then solve.

Mon Feb 28

Area meets circumference

TRY A circle's area and circumference have the same numerical value. Find the radius.

Tue Feb 29

Rings

TRY Two circles with the same center have radii 5 and 3. Find the area between them.

Wed Mar 1

Maximizing area

TRY A rectangle has perimeter 40. What is its largest possible area?

Thu Mar 2

Inscribed squares

TRY A square is inscribed in a circle of radius 6. Find the area of the square.

Fri Mar 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 27 • Mar 6–Mar 10, 2028 — Unit 6: Counting, Probability & Statistics

Counting Strategies **Core:** Multiply in stages; subtract what you do not want.

Mon Mar 6

Arrangements with a block

TRY In how many ways can 5 people line up if Al and Bo must stand together?

Tue Mar 7

Complementary counting

TRY How many 4-digit numbers have at least one repeated digit?

Wed Mar 8

Casework

TRY How many positive integers less than 100 have digits that add to 10?

Thu Mar 9

Repeated letters

TRY How many ways can the letters of BANANA be arranged?

Fri Mar 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 28 • Mar 13–Mar 17, 2028 — Unit 6: Counting, Probability & Statistics

Combinations & Pascal's Triangle **Core:** $\binom{n}{k}$ counts choices, and Pascal's triangle builds them.

Mon Mar 13

Choosing

TRY How many ways can you choose 3 pizza toppings from 10?

Tue Mar 14

Pascal's triangle

TRY What is the sum of the numbers in row 8 of Pascal's triangle? (Row 0 is just 1.)

Wed Mar 15

Binomial expansion

TRY Find the coefficient of x^2 in $(x+2)^5$.

Thu Mar 16

Stars & bars

TRY How many ways can 8 identical candies be given to 3 children (some may get none)?

Fri Mar 17

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 29 • Mar 20–Mar 24, 2028 — Unit 6: Counting, Probability & Statistics

Probability **Core:** Count carefully; multiply for "and," add for "or" when there is no overlap.

Mon Mar 20
Dice sums
TRY Two dice are rolled. What is the probability the sum is at least 10?

Tue Mar 21
Choosing a group
TRY Two people are chosen from 4 girls and 3 boys. P(one girl and one boy)?

Wed Mar 22
Complements
TRY Three dice are rolled. What is the probability of at least one 6?

Thu Mar 23
Geometric probability
TRY A point is picked at random inside a square. P(it lies inside the square's inscribed circle)?

Fri Mar 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Spring Break • Mar 27–Mar 31, 2028 • no class (not counted in the 34 weeks)

WEEK 30 • Apr 3–Apr 7, 2028 — Unit 6: Counting, Probability & Statistics

Expected Value & Statistics **Core:** Expected value is an average weighted by probability.

Mon Apr 3
Expected value
TRY What is the expected value of one roll of a fair die?

Tue Apr 4
Fair games?
TRY You win \$10 with probability $\frac{1}{5}$ and otherwise lose \$2. What are your expected winnings?

Wed Apr 5
Changing the mean
TRY The mean of 5 numbers is 12. Adding a sixth makes the mean 13. What is the sixth number?

Thu Apr 6
Weighted averages
TRY Class A (20 students) averages 80; class B (30 students) averages 90. Find the combined average.

Fri Apr 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 31 • Apr 10–Apr 14, 2028 — Unit 6: Counting, Probability & Statistics

Logic, Parity & Pigeonhole **Core:** Look for what cannot change, and plan for the worst case.

Mon Apr 10
Pigeonhole principle
TRY How many people do you need to be sure two were born in the same month?

Tue Apr 11
Parity
TRY Can five numbers chosen from $1, 3, 5, \ldots, 19$ add to 50?

Wed Apr 12
Invariants
TRY Write 1 to 10 on a board. Repeatedly erase two numbers and write their difference. Can the last number be 0?

Thu Apr 13
Truth-tellers & liars
TRY A says, "At least one of A and B is a liar." Each always tells the truth or always lies. Who is who?

Fri Apr 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 32 • Apr 17–Apr 21, 2028 — Contest Season Wrap-Up

Contest Strategy: AMC 8 & AMC 10 Preview **Core:** Choose the method that fits, then check with a second one.

Mon Apr 17
Estimate & simplify
TRY Compute $\sqrt{0.0049 \times 10^6}$.

Tue Apr 18
Test the choices
TRY Which makes $x^3 - 2x = 56$ true: 2, 3, 4 or 5?

Wed Apr 19
AMC 10 style
TRY If $3^x = 5$, find 9^{x+1}.

Thu Apr 20
Timed set: 10 in 20 min
TRY How many integers n satisfy $n^2 < 50$?

Fri Apr 21
Weekly Quiz
10 mixed contest problems, timed (25 min).

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WEEK 33 • Apr 24–Apr 28, 2028 — Contest Season Wrap-Up

Full Mock Contest & Review **Core:** Simulate contest day and learn from every error.

Mon Apr 24
Review: algebra
TRY If $2x + 3y = 13$ and $3x + 2y = 12$, find $x + y$.

Tue Apr 25
Review: geometry
TRY A right triangle has legs 7 and 24. Find the hypotenuse.

Wed Apr 26
Review: counting
TRY How many diagonals does a 10-sided polygon have?

Thu Apr 27
Full Mock Contest
Full-length mixed contest under real time limits.

Fri Apr 28
Error-log review
Redo every missed problem from the mock; write the key idea for each.

WEEK 34 • May 1–May 5, 2028 — Contest Season Wrap-Up

Final Contest & Celebration **Core:** Show what you have learned and set next year's goals.

Mon May 1
Light review: favorite strategies
TRY Find $1 + 2 + 3 + \cdots + 100$.

Tue May 2
Mixed warm-up set
TRY Solve $x^2 - 10x + 25 = 0$.

Wed May 3
Final Contest
Full-length contest covering the whole year.

Thu May 4
Contest review
Go over solutions together; students present one problem each.

Fri May 5
Celebration & reflection
Certificates, progress review, and goals for next year.

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ANSWER KEY

Answers to the Daily "Try" Problems

Week 1
Mon: 16
Tue: 19
Wed: 8
Thu: \$48

Week 2
Mon: $x = 14$
Tue: $x = 26$
Wed: $b = \frac{2A}{h} - c$
Thu: $k = 6$

Week 3
Mon: $x = 7$ or $x = -2$
Tue: $-1 < x < 7$
Wed: 11
Thu: 6

Week 4
Mon: After 2.5 hours
Tue: $\frac{10}{3}$ L ($3\frac{1}{3}$ L)
Wed: 12 hours
Thu: 10:30

Week 5
Mon: -2
Tue: x-int. 2, y-int. 6
Wed: $k = 13$
Thu: \$30/hour, \$80 fee

Week 6
Mon: $y = 2x - 7$
Tue: $y = -\frac{1}{2}x + 5$
Wed: 24
Thu: $2x + 3y = 12$

Week 7
Mon: $(4, 7)$
Tue: $(3, 2)$
Wed: 21
Thu: $a = 3$

Week 8
Mon: 47
Tue: 23
Wed: 2 km/h
Thu: 13

Week 9
Mon: $-2 < x \leq 4$
Tue: 4
Wed: 7
Thu: 15

Week 10
Mon: $x = 1$
Tue: $\frac{5}{2}$
Wed: 20
Thu: 1

Week 11
Mon: x^4
Tue: $\frac{11}{24}$
Wed: $x = \frac{9}{2}$
Thu: 2^{100} (since $32^{20} > 27^{20}$)

Week 12
Mon: $6\sqrt{5}$
Tue: $6\sqrt{2}$
Wed: 1
Thu: $2\sqrt{3}$

Week 13
Mon: 1.2×10^4
Tue: 12
Wed: 7
Thu: 12 and 13

Week 14
Mon: 400
Tue: 0 ($n(n+1)$ is always even)
Wed: 24
Thu: 30

Week 15
Mon: 6
Tue: 6
Wed: 4
Thu: 8

Week 16
Mon: $2x^2 + x - 15$
Tue: 29
Wed: 4
Thu: 999,999

Week 17
Mon: $3x^2(2x - 3)$
Tue: $(x - 8)(x + 3)$
Wed: $(2x + 1)(x + 3)$
Thu: 2: $(8, 7)$ and $(4, 1)$

Week 18
Mon: $x = 3$ or 4
Tue: $(x + 3)^2 - 7$
Wed: $x = 2 \pm \sqrt{5}$
Thu: 6

Week 19
Mon: Sum 9, product 14
Tue: 19
Wed: $k = 6$
Thu: $b = 4$, $c = -21$

Week 20
Mon: $5\sqrt{3}$
Tue: 1
Wed: $x = 7$ or -2
Thu: 23

Week 21
Mon: 14
Tue: -3
Wed: 10
Thu: 25

Week 22
Mon: 7
Tue: 192
Wed: 6 and 18
Thu: $x = -4$

Week 23
Mon: 860
Tue: 2,047
Wed: $\frac{10}{21}$
Thu: 2,500

Week 24
Mon: 10
Tue: 24
Wed: Center $(1, 1)$, radius 5
Thu: 30

Week 25
Mon: 45 ft
Tue: 52
Wed: 5 and $5\sqrt{3}$
Thu: 4.8

Week 26
Mon: 2
Tue: 16π
Wed: 100
Thu: 72

Week 27
Mon: 48
Tue: 4,464
Wed: 9
Thu: 60

Week 28
Mon: 120
Tue: 256
Wed: 80
Thu: 45

Week 29
Mon: $\frac{1}{6}$
Tue: $\frac{4}{7}$
Wed: $\frac{91}{216}$
Thu: $\frac{\pi}{4}$

Week 30
Mon: 3.5
Tue: \$0.40
Wed: 18
Thu: 86

Week 31
Mon: 13
Tue: No: five odd numbers have an odd sum
Wed: No: the sum always stays odd
Thu: A tells the truth; B lies

Week 32
Mon: 70
Tue: 4
Wed: 225
Thu: 15

Week 33
Mon: 5
Tue: 25
Wed: 35

Week 34
Mon: 5,050
Tue: $x = 5$

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